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Backend Development Training Overview

Course: ASP.NET Core MVC & Full Stack Development

Instructor: Eng. Yasmeeen Samir

Technologies: HTML, CSS, SQL Server, C#, OOP, ASP.NET Core MVC



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Submitted by

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ID: 2231039

Contents

1. Introduction

2. Training overview

2.1 Training Objectives

2.2 Training Contents

2.3 Learning Process

2.4 Importance of Practical Application

3. Key topics covered

3.1 Some tasks

4. Project showcase

4.1 Project title

4.2 Description



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4.3 Technologies Used

4.4 Key Features

4.5 Final Project

5. Key Learnings and Achievements

6. Future Goals and Aspirations

7. Conclusion

Introduction

As a fourth-year student at the Computers and Artificial Intelligence College at Sadat University, I'm Zeyad Mohamad Ammar pleased to present this training documentation to showcase my journey and accomplishments in backend development using .Net framework.

Throughout my academic career, I have developed a keen interest in the field of computer science, with a particular focus on backend engineering and its crucial role in modern web application development. This training opportunity has allowed me to deepen my understanding of server-side programming and gain hands-on experience in building robust and scalable web applications.

During this training, I have had the privilege of immersing myself in various aspects of backend development, including:



1. Proficient Programming in C#: I have acquired a strong foundation in the C# programming language, mastering its syntax, control structures, and programming best practices.
2. Expertise in the .Net Framework: I have extensively worked with the .Net framework, a popular C# web application framework, and have become adept at leveraging its features and tools to expedite the development process.
3. Efficient Database Management: I have developed skills in managing and interacting with databases, including the creation, manipulation, and querying of data using MS SQL and other database management systems.
4. Entity Framework (EF): I have Learned to Use EF to deal with w/ database in my backend code.
5. Practical Web Application Development: I have applied my knowledge of C# and .Net to develop several web applications, demonstrating my ability to translate requirements into functional and maintainable code.

This training has been instrumental in strengthening my technical skills and preparing me for a career in the field of computers and artificial intelligence. I am confident that the knowledge and experience I have gained will be invaluable in my future endeavors and will contribute to my continued growth and success as a computer science professional.



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I am excited to share the details of my training and the project I have undertaken, and I hope that this documentation will provide valuable insights into my progress and achievements.



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Training Overview

2026 Winter Code Camp on Software Development & Testing

Institution: FCAI USC

Training Duration: One Month (120 Hours)

Start Date: February

Mode of Delivery: Blended (combining online and in-person learning)

The 2026 winter Code Camp, Under the expert guidance of Eng. Yasmeeen Samir, was an intensive one-month training program designed for undergraduate students passionate about software development and testing. This program focused on building a solid foundation in developing web applications using .Net, emphasizing both the theoretical and practical aspects of web development. The training aimed to equip students with the essential skills and hands-on experience necessary to create dynamic, data-driven applications in a real-world context.



▪ Training Objectives

1. To develop fundamental skills in HTML, CSS, and JavaScript: Laying the groundwork for front-end development with a strong focus on responsive and interactive web design.
2. To build a foundational understanding of SQL databases: Enabling students to design, manage, and query relational databases effectively.
3. To introduce core concepts in C# programming: Providing a comprehensive guide to server-side scripting, allowing students to create dynamic web applications.
4. To introduce the .Net framework: Enhancing students' backend development skills through the structured MVC architecture of .Net.



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● **Training CONTENTS**

1. HTML, CSS, and JavaScript: Basics of web design and front-end development, covering structure, styling, and interactivity.
2. Database Fundamentals (MS SQL): Core principles of relational databases, SQL queries, and managing data effectively.
3. C# Programming: Server-side scripting, covering C# syntax, control structures, functions, and integration with databases.
4. **.Net: Introduction to the .Net framework, with a focus on building scalable web applications using MVC architecture.**



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• Learning Process

The training employed a blended learning approach, consisting of:

- Lectures: Providing theoretical knowledge and conceptual understanding of web development and database management.
- Practical Tasks: Hands-on exercises that reinforced the skills learned in lectures, allowing students to apply their knowledge in real scenarios.
- Project-Based Evaluation: At the end of the program, participants completed a project to demonstrate their understanding and application of the course material. This final project was designed to assess their grasp of both the theoretical and practical aspects of the curriculum.

• Importance of Practical Application

This training emphasized the importance of practical applications. By completing tasks and working on a final project, students gained real-world experience that will serve them well in their future careers in software development. The hands-on approach ensured that students left the program with not only knowledge but also confidence in their abilities to develop and test web applications.



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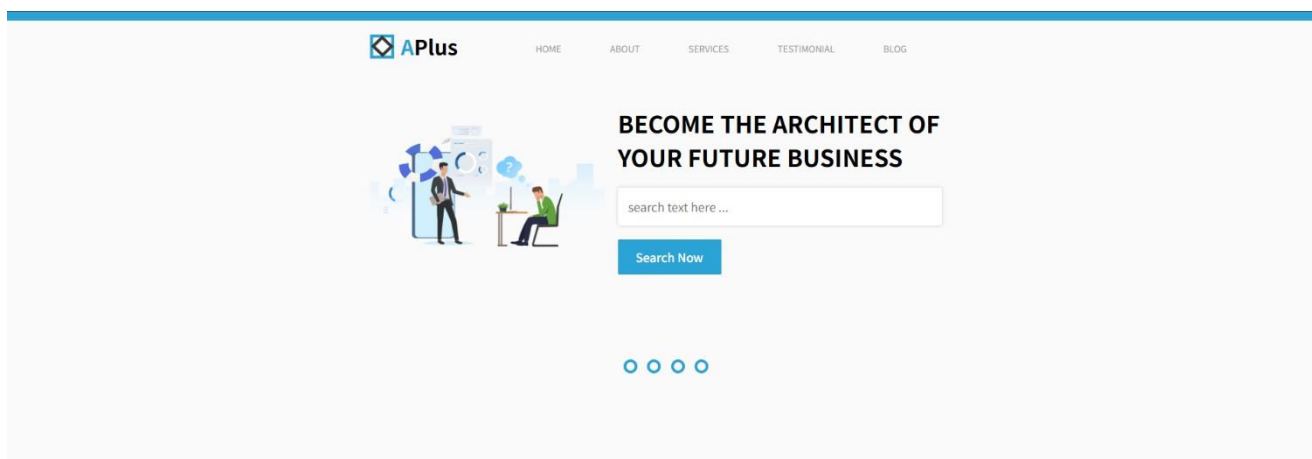
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Key Topics Covered

1. HTML

2. CSS & CSS3

Some of the tasks



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Our Services

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01

ADERU

It is a long established fact that a reader will be distracted by the readable content using Lorem Ipsum is that it has a

02

JOYMINI

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters,

03

JONSON

It is a long established fact that a reader will be distracted by the readable content using Lorem Ipsum is that it has a



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Let's talk about improving your business marketing...

It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters,

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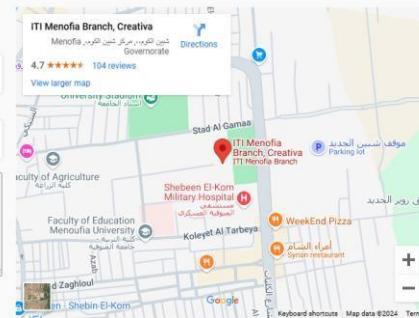
get in touch

name

Email

message

send



3. JavaScript

Some of the tasks

Id	Name	Actions
1	Leanne Graham	See All
2	Ervin Howell	See All
3	Clementine Bauch	See All
4	Patricia Lebsack	See All
5	Chelsey Dietrich	See All
6	Mrs. Dennis Schulist	See All
7	Kurtis Weissnat	See All
8	Nicholas Runolfsdottir V	See All
9	Glenna Reichert	See All
10	Clementina DuBuque	See All

Name: Leanne Graham

Email: Sincere@april.biz

Phone: 1-770-736-8031 x56442

Website: hildegard.org

Company: Romaguera-Crona

Address: Kulas Light, Gwenborough



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Name	
E-mail	
Password	
Gender	☐ Male ☐ Female
Sports	☐ Tennis ☐ Running ☐ Football
Country	
Submit Reset	

4. Database Management

- Introduction to SQL.
- Data manipulation using Entity Framework (EF).

5. C# Programming

- Basics of C# syntax and control structures.
- Object-oriented programming in C#.



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6. .Net Framework

- Installation and setup of .Net.
- Routes, controllers, and views.
- Middleware and service providers.

7. Web Application Development

- Building and deploying web applications.



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Project Showcase

Project Title: Manage Students in their Departments Website

Description: The Manage Students in their Departments Website is a .Net-based platform designed to manage students, departments, and administrators. It includes features such as user registration, added student, instructor. It also enables u to edit and delete them, which helped me to implement all CRUD operations. they are all enhanced with user-friendly styling and session handling for seamless navigation.

Technologies Used:

- .Net
- C#
- MS SQL
- HTML/CSS
- JavaScript



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Key Features:

- User authentication with sign-up, sign-in, and password editing capabilities, including validation and error handling.
- Full CRUD functionality for managing students, courses, assignments.
- Course enrollment features, allowing students to enroll in multiple courses.
- Admin dashboard for managing instructors, students, and courses.
- Responsive design with enhanced styling for a visually appealing experience.

Final project

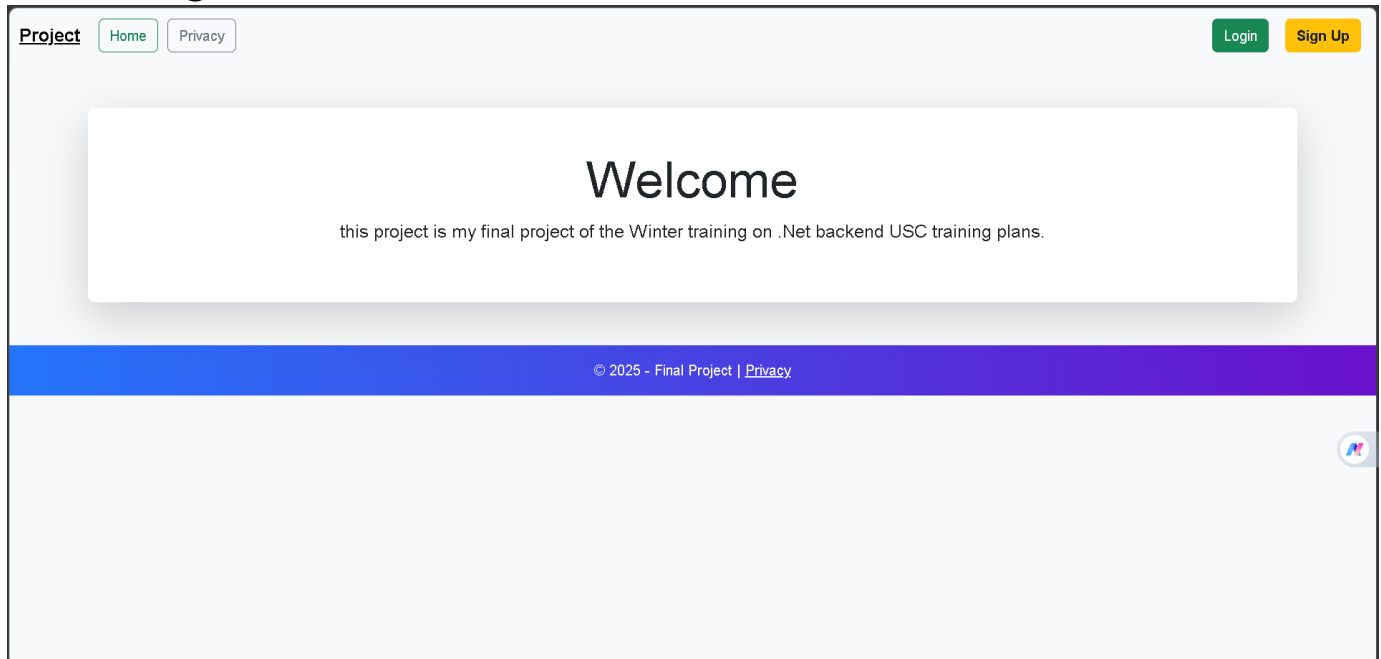


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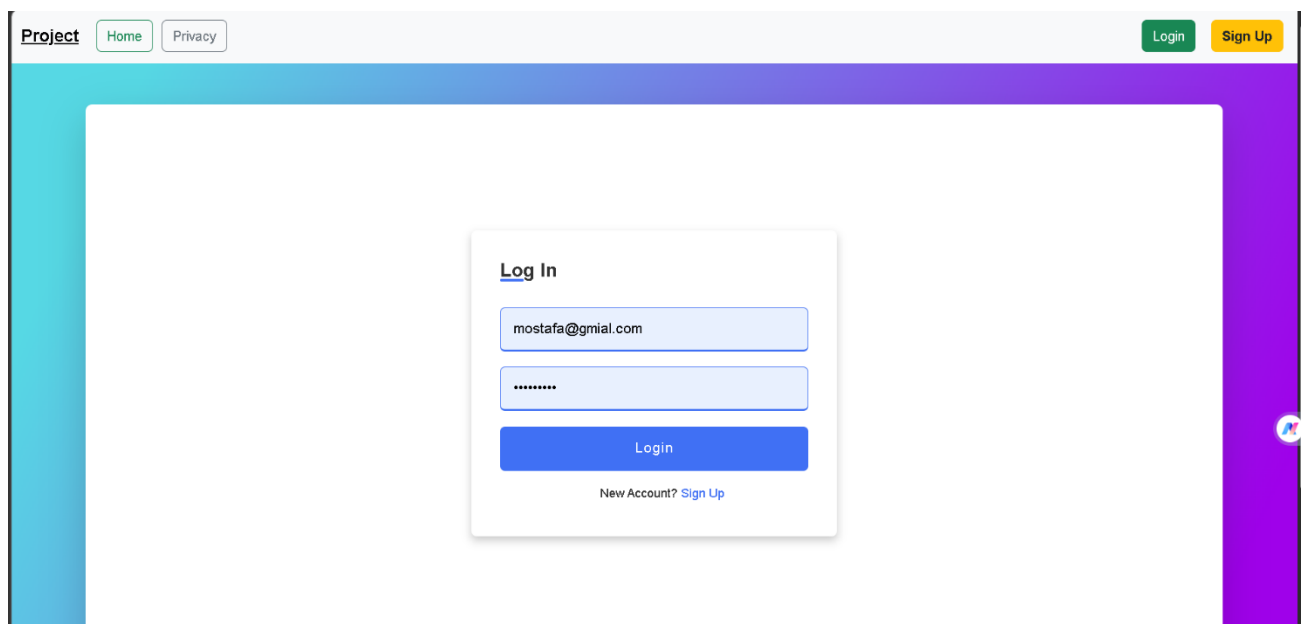


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Home Page.



Login page



Login validation



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Project Home Privacy Login Sign Up

Log In

Enter your email

The Email is Required

Enter your password

The Password is Required

Login

New Account? [Sign Up](#)

Register page.

Project Home Privacy Login Sign Up

Registration

Enter your name

Enter your email

Create password

Confirm password

Register Now

Already have an account? [Login now](#)



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Privacy Page.



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Students page

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All Students

[Add New Student](#)

Student ID	Student Name	Student Age	Student Address	Student Phone	Student Email	Department Name	
13	Zeyad Ammar	22	Kom-hamada	012323442434	Zeyadammar@gmail.com	MI	Details Edit Delete
14	Mohamed Salah	19	Monofia	01231213132	mohamed@yahoo.com	AI	Details Edit Delete
15	Mostafa Ammar	22	Kom-hamada	01145258396	mostafa@gmail.com	AI	Details Edit Delete

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Add New Student page



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Add Student

Student Name

Student Age

Student Address

Student Email

Student Phone

Password

ConfirmPassword

Student DepartmentID
—choose Ur Department——

Add Cancel

Edit Student Info Page.

Project Home Privacy Students Departments Mostafa Ammar Logout

Edit Student Zeyad Ammar

Student Name
Zeyad Ammar

Student Age
22

Student Address
Kom-hamada

Student Email
Zeyadammar@gmail.com

Student Phone
012323442434

Password
.....

ConfirmPassword
.....

Student DepartmentID
MI

Submit Cancel



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Details of Student page.



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Project Home Privacy Students Departments

To exit full screen, move mouse to top of screen or press and hold Esc

Details of Zeyad Ammar

Student Name:	Zeyad Ammar
Student Age:	22
Student Address:	Kom-hamada
Student Email:	Zeyadamm@gmail.com
Student Phone:	012323442434
Department Name:	MI

Cancel

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Delete Student page

Project Home Privacy Students Departments

Mostafa Ammar Logout

Sure To Delete Student Zeyad Ammar

Student Name:	Zeyad Ammar
Student Age:	22
Student Address:	Kom-hamada
Student Email:	Zeyadamm@gmail.com
Student Phone:	012323442434
Department Name:	MI

Delete

Cancel

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Departments Page

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All Departments

[Add New Department](#)

Id	Department Name	
4	BI	Details Edit Delete
5	MI	Details Edit Delete
13	cs	Details Edit Delete
16	IS	Details Edit Delete
17	AI	Details Edit Delete
19	sw	Details Edit Delete

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Add Department

Department Name

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Edit Department page

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Edit Department Bi

Department Name

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Details of Department Page.

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Details of AI

Department Name:
AI

Number Of Students:
2

The Department Students

Id	Name	Age	Address	Phone	Email	Department.Name	
14	Mohamed Salah	19	Monofia	01231213132	mohamed@yahoo.com	AI	Details Edit Delete
15	Mostafa Ammar	22	Kom-hamada	01145258396	mostafa@gmail.com	AI	Details Edit Delete

[Cancel](#)

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Mostafa Ammar [Logout](#)

Sure To Delete Department AI

Department Name:

AI

Delete

Cancel

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Login Admin Details.

[Project](#) [Home](#) [Privacy](#) [Students](#) [Departments](#)

Mostafa Ammar [Logout](#)

Details of Mostafa Ammar

Admin Name:

Mostafa Ammar

Admin Email:

mostafa@gmail.com

Cancel Edit

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Edit Admin Page.

Project Home Privacy Students Departments

Mostafa Ammar Logout

Edit Admin Mostafa Ammar

Admin Name
Mostafa Ammar

Admin Email
mostafa@gmail.com

Password

ConfirmPassword

Submit Cancel

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Logout Button Result

Project Home Privacy Students Departments

localhost:5177 says
are you sure to Log out Mostafa Ammar
OK Cancel

Mostafa Ammar Logout

Details of Mostafa Ammar

Admin Name:
Mostafa Ammar

Admin Email:
mostafa@gmail.com

Cancel Edit

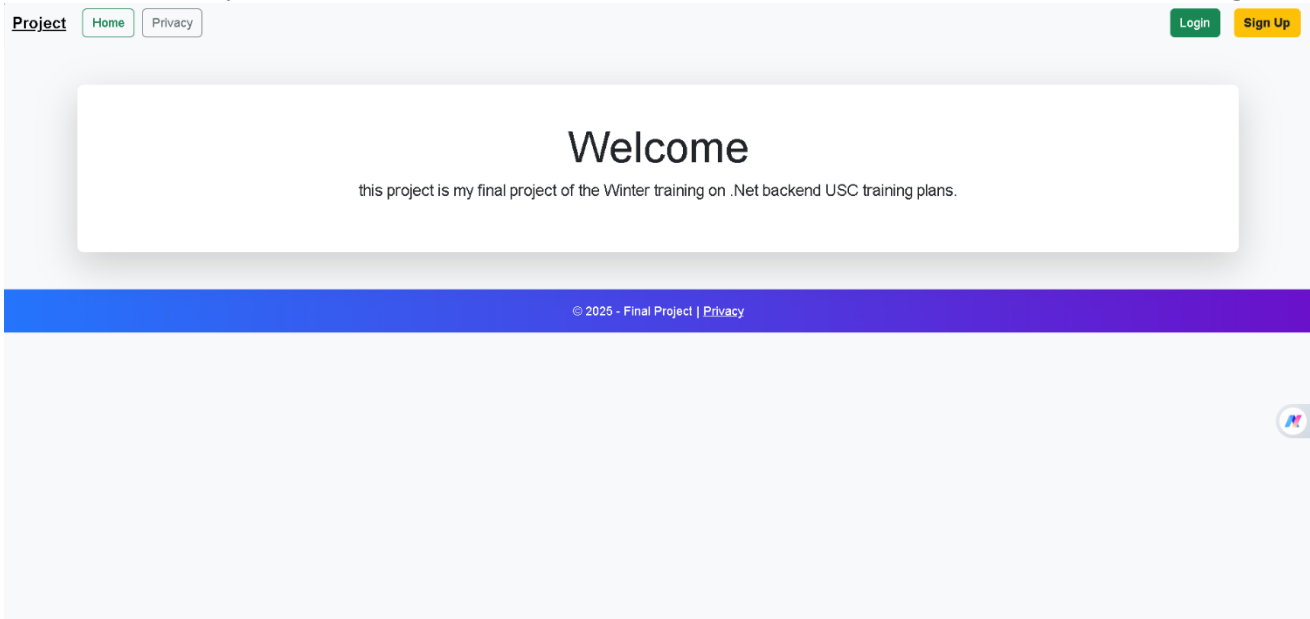
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Key Learnings and Achievements

- **.Net Proficiency:** Gained hands-on experience with .Net, enhancing my skills in building secure, dynamic web applications with effective session handling, database interactions, and routing.
- **CRUD Functionality Implementation:** Mastered the implementation of CRUD operations, enabling the smooth management of students, and courses. This was essential in creating a user-friendly admin dashboard for efficient data management.



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- **Database Management and Relationships:** Developed an understanding of database schema design and relationships, including one-to-many and pivot tables, to support complex data interactions like course enrollment and instructor assignments.
- **User Authentication and Validation:** Implemented secure user authentication with validation and error handling, improving both the user experience and the overall security of the application.
- **Front-End Design and Styling:** Improved front-end design skills, focusing on responsive, visually appealing layouts for enhanced user experience. Styled elements like buttons, cards, and sections to create an intuitive, cohesive interface.
- **Teamwork and Project Documentation:** Worked on preparing clear documentation and a presentation for the *Courses Website* project, which strengthened my skills in effectively communicating project details and collaborating in a team setting.
- **Adaptability and Problem Solving:** Developed problem-solving skills by troubleshooting various technical challenges, from database



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migrations to session handling, which sharpened my ability to adapt and find solutions quickly.

These key learnings have significantly improved my technical and soft skills, positioning me well for future academic and professional opportunities in software development and artificial intelligence.

Future Goals and Aspirations

As a fourth-year student at the Faculty of Computers and Artificial Intelligence at Sadat University, my goals are centered on deepening my knowledge in artificial intelligence and software development to make impactful contributions in these fields. I am particularly interested in areas like machine learning, data science, and advanced web development, with the aim of building intelligent systems that solve real-world problems efficiently.

In the future, I aspire to work on projects that integrate AI into everyday applications, making technology more intuitive and accessible. I also plan to further my studies by pursuing certifications or even a master's degree in



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artificial intelligence, focusing on areas like natural language processing or computer vision. This knowledge will allow me to develop innovative solutions that bridge the gap between complex AI models and practical applications.

The skills and experience I've gained through working on the Courses Website project—particularly in areas like database management, session handling, and CRUD operations in .Net—have given me a solid foundation in backend development and user-centered design. These skills are critical for future projects and will support my growth in building more advanced systems that require seamless user interaction and robust functionality.

This project has reinforced my interest in creating scalable, user-friendly applications, and I am excited to apply these principles as I continue to explore new areas in computer science and artificial intelligence.



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CONCLUSION

In conclusion, this training has been invaluable in strengthening my foundation in backend development and database management, with a focus on creating dynamic, user-centered applications. The experience has prepared me to tackle more complex projects, both academically and professionally, as I continue my journey in the field of computers and artificial intelligence. I am grateful for the opportunity to expand my skill set and grow through hands on learning and practical application, and I look forward to applying these skills in future endeavors.